

Optimizing Data for Decisions – Linear Case

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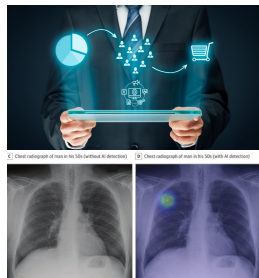
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Optimizing Datasets

We are increasingly living in a world shaped by data:

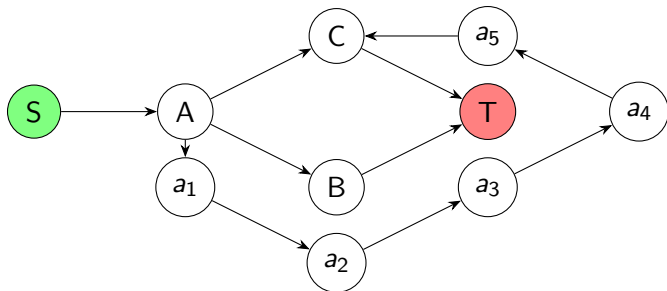
- Various personalized services and products based on data.
- Use of data to solve hard societal problems (detect cancers, predict pandemics, etc.)
- Everybody is convinced data is valuable. But **what type of data and how much data do we need for specific tasks?**
- This critical question is rarely asked (see below for exceptions), but will become increasingly vital as data continues to become the bedrock of our economy and society.
- In this paper, we aim to understand the **informational value (or informativeness)** of a given dataset to recover (or estimate) the optimal solution to a well-posed decision task—one that can be formulated as a linear optimization problem.



Data Informativeness in a Toy Example

- Consider a shortest path problem with **source S** and **target T**.
- Suppose the edge costs $c \in \mathbb{R}^{11}$ are unknown—**Data corresponds to edge costs**.

Which data informs the optimal decision here?

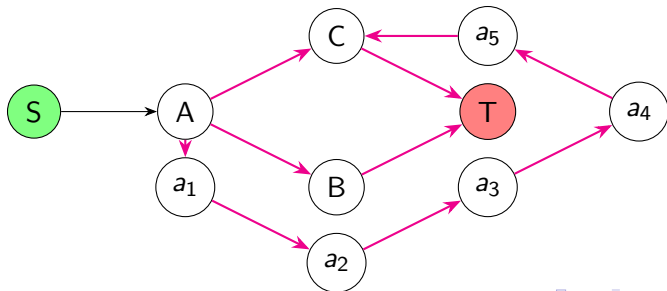


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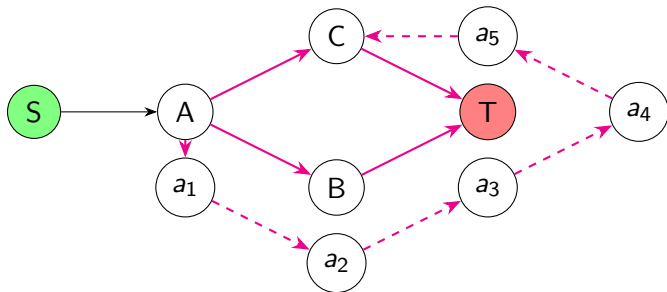
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The Role of Prior Information

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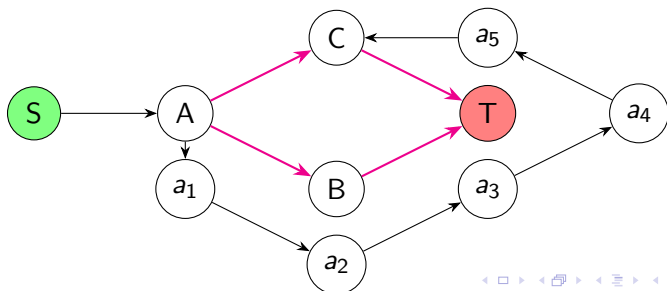
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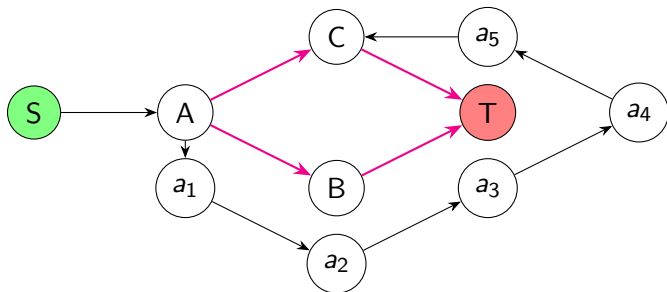
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- If this were a map on scale, we intuitively want to say that we don't need the "detour" $a_1, \dots, a_5$ as it is likely always a suboptimal choice.
- **With prior information** $c \in \mathcal{C} := [1, 5]^{11}$: we only need c_{AC}, c_{CT} and c_{AB}, c_{BT} . **Prior information is important!**



The Role of Prior Information

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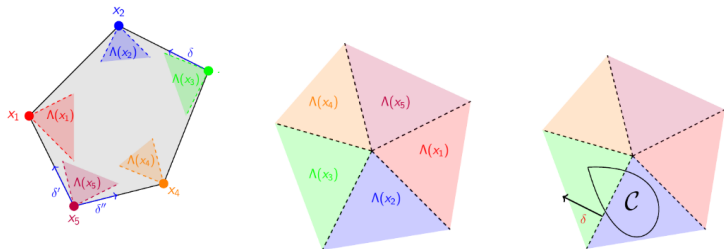
- With prior information $c \in \mathcal{C} := [1, 5]^{11}$: we only need c_{AC}, c_{CT} and c_{AB}, c_{BT} . **Prior information is important!**
- In fact, we only need the information $c_{AC} + c_{CT}$ and $c_{AB} + c_{BT}$.
- In practice, prior information comes from features such as edge costs being correlated with length, congestion, speed limit, historical travel time etc...



Our Results

- We focus on tasks that can be formulated as linear optimization problems, with unknown cost parameter and a polyhedral decision set.
- We prove a **necessary and sufficient condition** under which a dataset is sufficient to recover the optimal decision for a linear program under cost uncertainty.
 - Our condition enables incorporating prior information about the unknown parameter, encoded in an uncertainty set.
- We provide an alternative characterization in terms of reachable optimal solutions that enables algorithmically accessible formulation for evaluating and constructing sufficient datasets.
- Building on these characterizations, we develop an **iterative algorithm that constructs a minimal sufficient dataset**.

Intuition: A Few Query Directions Can Be Enough



- Solving the LP with cost vector c does *not* require learning the full vector; it only requires enough information to choose an optimizer.
- In the pictured prior region, only two neighboring optimality regions can occur, separated by the boundary $c^\top \delta = 0$.
- The scalar query $c^\top \delta$ tells which side of the boundary c lies on, or whether it lies on the boundary. Hence one observation recovers a valid optimal solution.

Background: Related Literature

- **Model compression for LP:** This line of work asks how to replace a large LP by a lower-dimensional or sketched formulation while approximately preserving the geometry that matters for optimization.

[Vu18; Poirion23; Sakaue–Oki24]

- **Data-driven algorithm design:** The algorithm has tunable parameters. Samples from an instance distribution are used to choose these parameters, and the guarantee concerns future performance of the learned algorithm.

[Gupta–Roughgarden17/20; Balcan et al. 17/18/21/24]

- **Sample compression scheme:** A small subset of training examples certifies the learned rule. When this certificate is stable, realizability yields fast distribution-free generalization bounds.

[Valiant84; Littlestone–Warmuth86; Hanneke–Kontorovich21]

- **Active learning:** Instead of observing all information up front, the learner asks targeted queries, often only where the current candidate decisions disagree.

[Dasgupta11; Hanneke14; Balcan et al. 06/09]

Our viewpoint. In LPs, we learn linear measurements of the unknown cost that are sufficient for exact optimal-decision recovery.

[Bennouna et al. 25/26; Ye26]

Problem Formulation

- **Decision-task:** a linear program $\min_{x \in \mathcal{X}} c^\top x$, where $\mathcal{X} \subset \mathbb{R}^d$ is a bounded polyhedron.
- **Prior information:** The cost vector c is unknown. Prior information is modeled through an **uncertainty set** $c \in \mathcal{C} \subset \mathbb{R}^d$. E.g.

$$\mathcal{C} = \{c \in \mathbb{R}^d : \exists \alpha \in [\underline{\alpha}, \bar{\alpha}], \exists \epsilon \in [-\eta, \eta]^d, c = \alpha^\top \Phi + \epsilon\}$$

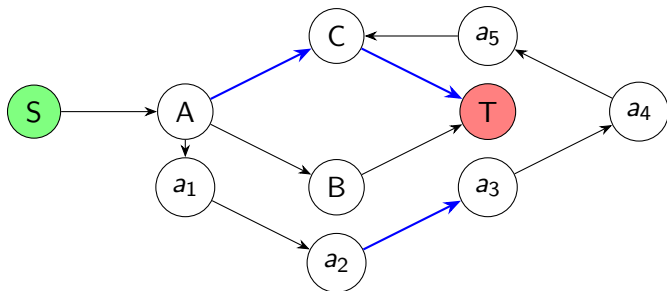
for some features $\Phi \in \mathbb{R}^{k \times d}$.

- **Dataset:** Consists of a set of queries $\mathcal{D} = \{q_1, \dots, q_N\} \subset \mathbb{R}^d$, which provides the observations $c^\top q_1, \dots, c^\top q_N$.
 - Allows to measure linear combinations of unknown cost values.
 - Focus on noiseless case in this talk.

Problem Formulation

- **Decision-task:** a linear program $\min_{x \in \mathcal{X}} c^\top x$, where $\mathcal{X}$ is a polyhedron.
- **Prior information:** We know that $c \in \mathcal{C}$.
- **Dataset:** $\mathcal{D} = \{q_1, \dots, q_N\} \subset \mathbb{R}^d$ provides the observations $c^\top q_1, \dots, c^\top q_N$.

Example: $\mathcal{D} = \{\delta_{AC} + \delta_{CT}, \delta_{a_2 a_3}\}$, where δ_e is the indicator vector at edge e , provides observations $\{c_{AC} + c_{CT}, c_{a_2 a_3}\}$.



Problem Formulation: Sufficient Datasets

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Given a *decision task* (defined by $\mathcal{X}$) and *prior information* (defined by $\mathcal{C}$), what *dataset* enables finding the task-optimal decision?

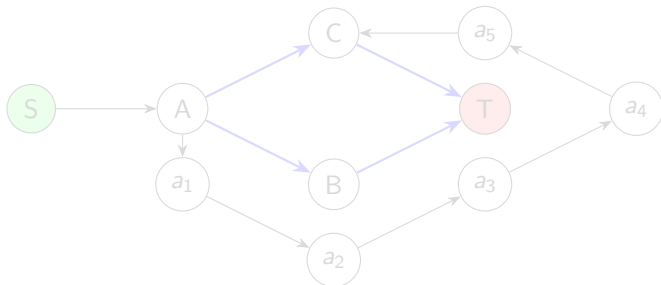
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Definition (Sufficient Dataset)

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$$\forall c \in \mathcal{C}, \quad \hat{x}(c^\top q_1, \dots, c^\top q_N) \in \arg \min_{x \in \mathcal{X}} c^\top x.$$

$\mathcal{D} = \{\delta_{AC} + \delta_{CT}, \delta_{AB} + \delta_{BT}\}$ is sufficient for $\mathcal{C} = [1, 5]^{11}$ in the example below.



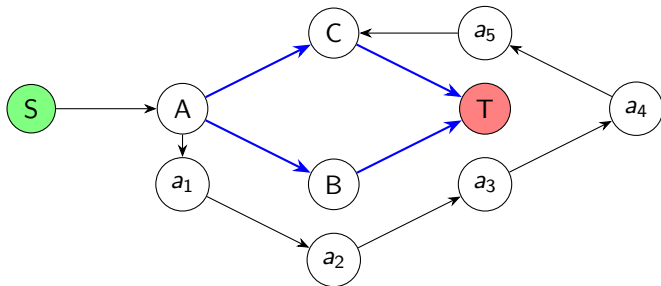
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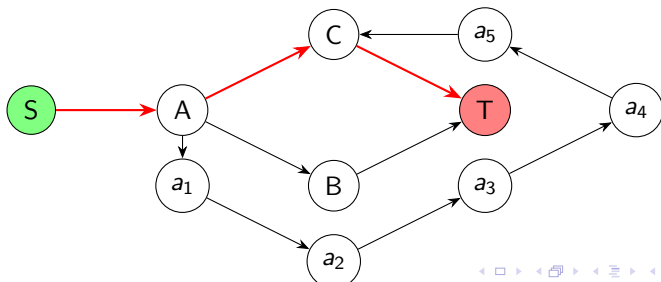
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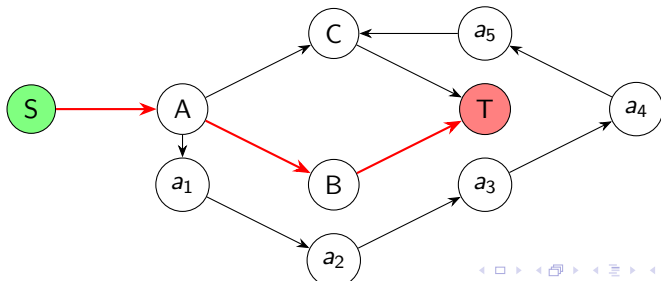
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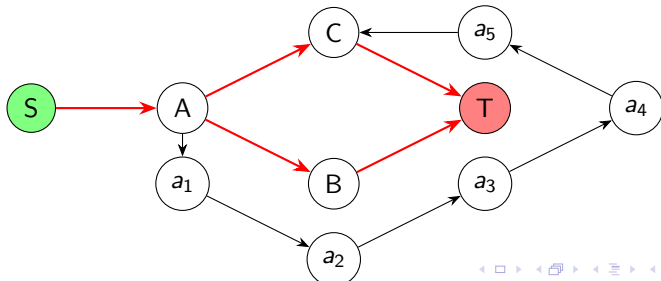
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Characterizing Sufficient Datasets

Useful Characterization for Sufficient Datasets

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$$\forall c \in \mathcal{C}, \quad \hat{x}(c^\top q_1, \dots, c^\top q_N) \in \arg \min_{x \in \mathcal{X}} c^\top x.$$

- Notice that observing $c^\top q$ for all $q \in \mathcal{D}$ is equivalent to observing the projection of c onto the span of $\mathcal{D}$.

Proposition

$\mathcal{D} := \{q_1, \dots, q_N\}$ is sufficient for uncertainty set $\mathcal{C}$ and decision set $\mathcal{X}$ if and only if

$$\forall c, c' \in \mathcal{C}, \quad c_{\text{span } \mathcal{D}} = c'_{\text{span } \mathcal{D}} \implies \arg \min_{x \in \mathcal{X}} c^\top x = \arg \min_{x \in \mathcal{X}} c'^\top x.$$

Warm-up: Characterization Under No Prior Information

Let us write $\mathcal{X} = \{x \in \mathbb{R}^d : Ax = b, x \geq 0\}$.

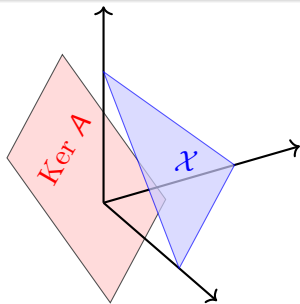
Proposition

Suppose $\mathcal{C} = \mathbb{R}^d$ (no prior information). $\mathcal{D}$ is a sufficient dataset if and only if

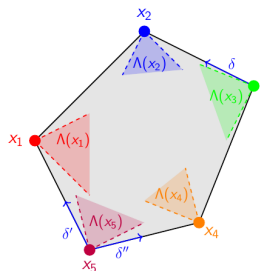
$$\text{Ker } A \subset \text{span } \mathcal{D}$$

Intuition:

- Can decompose the objective function as $c_{\text{Ker } A}^\top x + c_{(\text{Ker } A)^\perp}^\top x$.
- Any change from a feasible x to feasible $x + \delta$ satisfies $\delta \in \text{Ker } A$.
- Hence, $c_{(\text{Ker } A)^\perp}^\top x$ is constant across all feasible solutions.



Linear Programming Basics

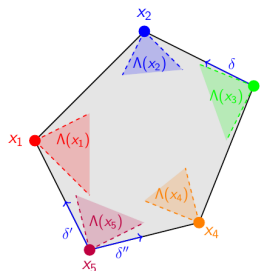


Extreme Points ($\mathcal{X}^\angle$). $x \in \mathcal{X}$ with no $\lambda \in (0, 1)$ and $y, z \in \mathcal{X}$ such that $x = \lambda y + (1 - \lambda)z$.

Extreme Directions ($D(x^*)$). For every $x^* \in \mathcal{X}^\angle$, its extreme directions are feasible directions ($\delta \in \mathbb{R}^d : \exists \varepsilon > 0, x^* + \varepsilon \delta \in \mathcal{X}$) that cannot be written as a convex combination of two non-proportional feasible directions.

Optimality Cones ($\Lambda(x^*)$). Every $x^* \in \mathcal{X}^\angle$ is associated with a cone $\Lambda(x^*) = \{c \in \mathbb{R}^d : x^* \in \arg \min_{x \in \mathcal{X}} c^\top x\}$.

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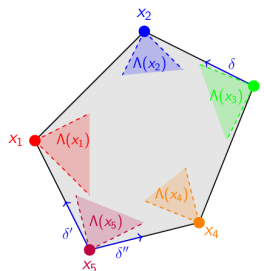


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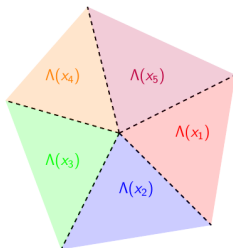
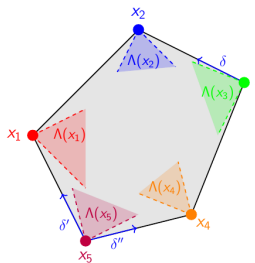


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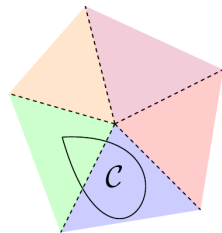
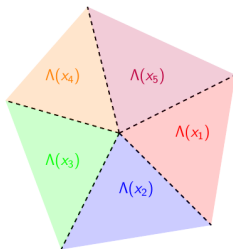
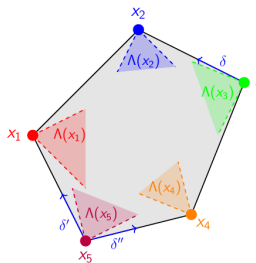
Optimality Cones



x_i Extr Points
 $\Lambda(x^*)$ Opt Cones
 $\delta, \delta', \delta''$ Extr Directions

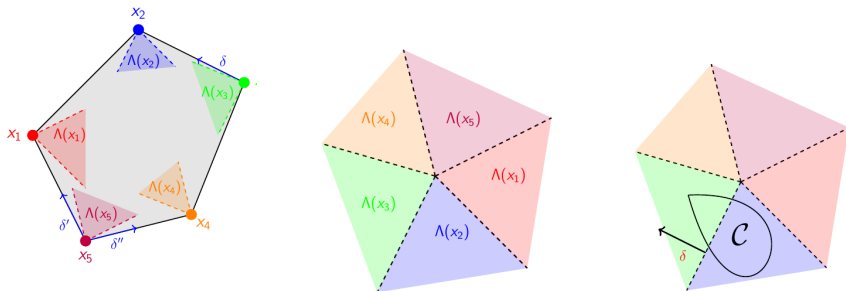
- **Optimality Cones.** For x^* extr point, $\Lambda(x^*) = \{c : x^* \in \arg \min_{x \in \mathcal{X}} c^\top x\}$.
- The cones form a partition: $\bigcup_{x^* \in \mathcal{X}^*} \Lambda(x^*) = \mathbb{R}^d$.
- $c \in \Lambda(x^*) \iff x^*$ is optimal for cost c .
- Solving an LP with cost $c \iff$ finding which cone $\Lambda(x^*)$ c belongs to.

A Geometric Perspective on Dataset Sufficiency



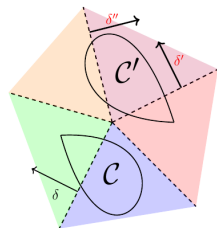
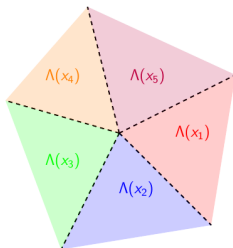
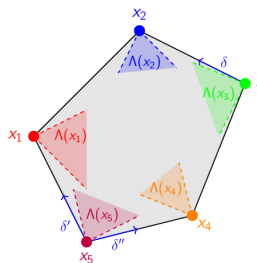
- Back to our problem: A sufficient dataset $\mathcal{D}$ needs therefore to *locate* c with respect to the cones, using the projections $c^\top q, q \in \mathcal{D}$.
- Given an uncertainty set $\mathcal{C}$: A sufficient dataset needs to discriminate the **remaining possible cones** in $\mathcal{C}$ — $\Lambda(x_2)$ and $\Lambda(x_3)$ in the example.

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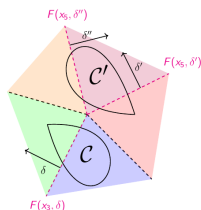
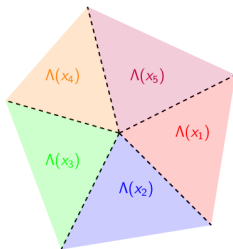
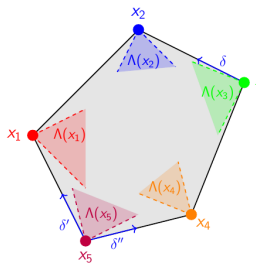
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- Given an uncertainty set $\mathcal{C}$: A sufficient dataset needs to discriminate the **remaining possible cones** in $\mathcal{C}$ — $\Lambda(x_2)$ and $\Lambda(x_3)$ in the example.
- In the example, it suffices to observe the projection $c^\top \delta$ which discriminates the cones $\Lambda(x_2)$ and $\Lambda(x_3)$.

A Geometric Perspective on Dataset Sufficiency



- For the set $\mathcal{C}$, observing the projection $c^\top \delta$ allows discriminating the cones $\Lambda(x_2)$ and $\Lambda(x_3)$.
- For the set $\mathcal{C}'$, observing $c^\top \delta'$ and $c^\top \delta''$ allow discriminating between $\Lambda(x_4)$, $\Lambda(x_5)$, $\Lambda(x_1)$.

A Geometric Perspective on Dataset Sufficiency



- More generally: each extr direction δ induces a face of a cone $F(x^*, \delta)$.
- A sufficient dataset needs to span extr directions δ of faces $F(x^*, \delta)$ intersecting the uncertainty set $\mathcal{C}$.

Definition (Relevant Extreme Directions)

$$\Delta(\mathcal{X}, \mathcal{C}) = \{\delta \in \mathbb{R}^d : \exists x^* \in \mathcal{X}^{\angle}, \delta \in D(x^*) \text{ and } F(x^*, \delta) \cap \mathcal{C} \neq \emptyset\}.$$

Exact Characterization

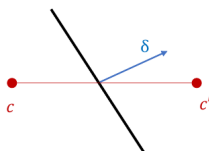
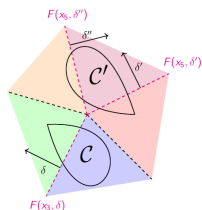
Theorem

Assume $\mathcal{C}$ convex open.

$\mathcal{D}$ is sufficient $\iff \Delta(\mathcal{X}, \mathcal{C}) \subset \text{span } \mathcal{D}$.

Proof Sketch:

- Show $\Delta(\mathcal{X}, \mathcal{C}) \subset \text{span } \mathcal{D} \rightarrow \mathcal{D}$ is sufficient.
- Suppose $\mathcal{D}$ is not sufficient. Then, $\exists c, c' \in \mathcal{C}$ such that $c_{\text{span } \mathcal{D}} = c'_{\text{span } \mathcal{D}}$, but the corresponding arg min's are different ($\rightarrow c, c'$ in different cones).
- We have $c - c' \in (\text{span } \mathcal{D})^\perp$ and $\delta \in \Delta(\mathcal{X}, \mathcal{C})$.
- $(\text{span } \mathcal{D})^\perp \subset \Delta(\mathcal{X}, \mathcal{C})^\perp$ only if $c - c' \perp \delta$, which can only happen if c and c' on the face, contradiction.

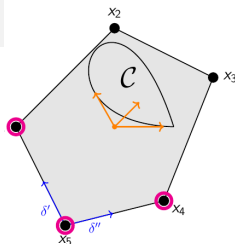


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- Provides an exact geometric characterization.
- Downside: Hard to compute! It requires finding all possible optimal solutions that are “neighbors”.

$$\Delta(\mathcal{X}, \mathcal{C}) = \{x - x' : \exists c \in \mathcal{C}, x, x' \in \operatorname{argmin}_{x \in \mathcal{X}} c^\top x \cap \mathcal{X}^\angle\}$$

A Tractable Characterization

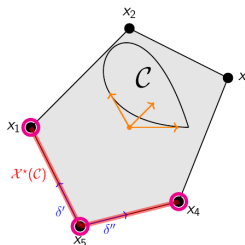
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- If we relax the “neighboring condition”, we get a set of *reachable solutions* $\mathcal{X}^*(\mathcal{C}) := \bigcup_{c \in \mathcal{C}} \underset{x \in \mathcal{X}}{\text{argmin}} c^\top x$.



A Tractable Characterization

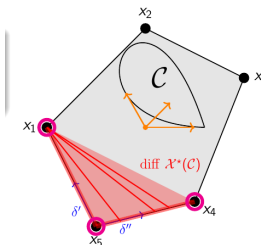
Theorem

Assume $\mathcal{C}$ convex open.

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- Its associated differences $\text{diff } \mathcal{X}^*(\mathcal{C}) = \{x - x' : x, x' \in \mathcal{X}^*(\mathcal{C})\}$.



Theorem

Assume $\mathcal{C}$ is convex. We have $\text{span } \Delta(\mathcal{X}, \mathcal{C}) = \text{span } \text{diff } \mathcal{X}^*(\mathcal{C})$.

A Tractable Characterization

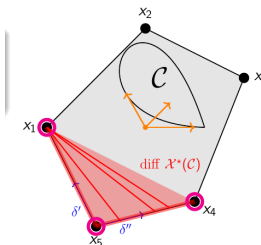
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A Tractable Characterization

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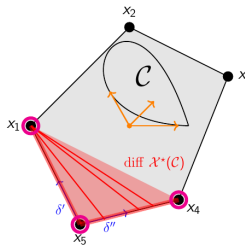
We have $\text{span } \Delta(\mathcal{X}, \mathcal{C}) = \text{span } \text{diff } \mathcal{X}^*(\mathcal{C})$.

Combining both results gives:

Theorem

Assume $\mathcal{C}$ convex open.

$\mathcal{D}$ is sufficient $\iff \text{span } \text{diff } \mathcal{X}^*(\mathcal{C}) \subset \text{span } \mathcal{D}$.



*Given a task ($\mathcal{X}$) and prior information ($\mathcal{C}$),
how to construct the **smallest** sufficient dataset?*

A Data Selection Algorithm: Smallest Sufficient Dataset

Recall $\mathcal{X}^*(\mathcal{C}) := \bigcup_{c \in \mathcal{C}} \operatorname{argmin}_{x \in \mathcal{X}} c^\top x$ and the characterization:

$$\mathcal{D} \text{ is sufficient} \iff \operatorname{span} \operatorname{diff} \mathcal{X}^*(\mathcal{C}) \subset \operatorname{span} \mathcal{D}$$

We can write $\operatorname{span} \operatorname{diff} \mathcal{X}^*(\mathcal{C}) = \operatorname{span} \{x^* - x_0 : x^* \in \mathcal{X}^*(\mathcal{C})\}$ for some fixed $x_0 \in \mathcal{X}^*(\mathcal{C})$.

① Start with $\mathcal{D} \leftarrow \emptyset$.

② Is there a *cost scenario* that changes *my decision* in a way NOT measured by my *data*?

Formally: Check if $\exists c \in \mathcal{C}$, $x^* \in \operatorname{argmin}_{x \in \mathcal{X}} c^\top x$ with $x^* - x_0 \notin \operatorname{span} \mathcal{D}$.

- If YES, add a measurement of their difference: $\mathcal{D} \leftarrow \mathcal{D} \cup \{x^* - x_0\}$.
- If NO, the data $\mathcal{D}$ is sufficient.

A Data Selection Algorithm: Detecting a Missing Direction

Key algorithmic step. Given the current dataset $\mathcal{D}$, check whether $\text{span } \mathcal{D}$ misses a reachable decision direction:

$$\exists c \in \mathcal{C}, \quad x^* \in \underset{x \in \mathcal{X}}{\operatorname{argmin}} c^\top x \quad \text{with} \quad x^* - x_0 \notin \text{span } \mathcal{D}.$$

Let $P_{\mathcal{D}^\perp}$ denote projection onto $(\text{span } \mathcal{D})^\perp$. Equivalently, test whether the norm problem has a positive optimum:

$$\sup_{c, x^*} \|P_{\mathcal{D}^\perp}(x^* - x_0)\| \quad \text{s.t.} \quad c \in \mathcal{C}, \quad x^* \in \underset{x \in \mathcal{X}}{\operatorname{argmin}} c^\top x.$$

- **Randomization trick.** Draw $\alpha \sim \mathcal{N}(0, I_d)$ once. For any fixed $v \neq 0$, $\Pr\{\alpha^\top v \neq 0\} = 1$.
- Thus, with probability one, it suffices to solve the two linearized search problems

$$\sup_{c, x^*} \alpha^\top P_{\mathcal{D}^\perp}(x^* - x_0) \quad \text{and} \quad \inf_{c, x^*} \alpha^\top P_{\mathcal{D}^\perp}(x^* - x_0) \quad \text{s.t.} \quad c \in \mathcal{C}, \quad x^* \in \underset{x \in \mathcal{X}}{\operatorname{argmin}} c^\top x.$$
- Both signs are needed: $\alpha^\top v$ may be positive or negative. Since $x_0 \in \mathcal{X}^*(\mathcal{C})$, value 0 is feasible; append the returned direction if the supremum is > 0 or the infimum is < 0 .

A Data Selection Algorithm: MIP Implementation

In either linearized search problem, the key constraint is $x^* \in \operatorname{argmin}_{x \in \mathcal{X}} c^\top x$. For $\mathcal{X} = \{x \geq 0 : Ax = b\}$, LP optimality is equivalent to primal-dual feasibility plus complementary slackness:

$$x^* \geq 0, \quad s \geq 0, \quad Ax^* = b, \quad A^\top \lambda + s = c, \quad x_i^* s_i = 0 \quad \forall i.$$

- Together with $c \in \mathcal{C}$, this gives a linear objective over (c, x^*, λ, s) , plus complementarity.
- Complementarity is encoded with binary variables, e.g., $0 \leq x_i^* \leq M_i z_i$ and $0 \leq s_i \leq \bar{M}_i(1 - z_i)$.
- When $\mathcal{C}$ is polyhedral and valid bounds are available, each search call is a mixed-integer program with $O(d + \#\text{constr}(\mathcal{X}))$ variables and $O(d + \#\text{constr}(\mathcal{X}) + \#\text{constr}(\mathcal{C}))$ constraints.

Takeaway. This gives an implementable MIP routine for the data-selection algorithm. It also motivates the hardness question: can we prove NP/coNP-hardness results for finding SDDs? We will address this in the second half of the talk.

Applications

Application: Shortest Path Problems

- The task is to navigate from the **origin** to **destination** in a neighborhood of Cambridge, MA.
- Assume we have an estimation $c_0 > 0$ of the edges travel time.
- Travel times (costs) c are uncertain and vary in:
 $(1 - \varepsilon)c_0 \leq c \leq (1 + \varepsilon)c_0$,
where $\varepsilon \in [0, 1]$.
- Extreme directions δ in this problem are **cycles**.
- Applications: Railroad planning, navigation in altered/damaged networks...

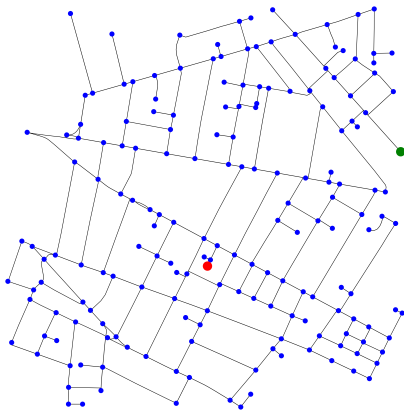


Figure: Neighborhood in Cambridge

Application: Shortest Path Problems

- $c \in \mathcal{C}_\varepsilon := \{c \in \mathbb{R}^d, (1 - \varepsilon)c_0 \leq c \leq (1 + \varepsilon)c_0\}$.
- When $\varepsilon = 0$, $c = c_0$.

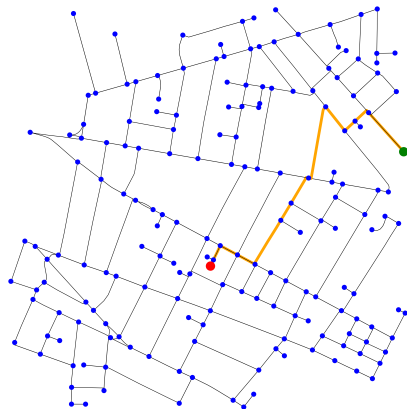


Figure: Shortest path for $c = c_0$.

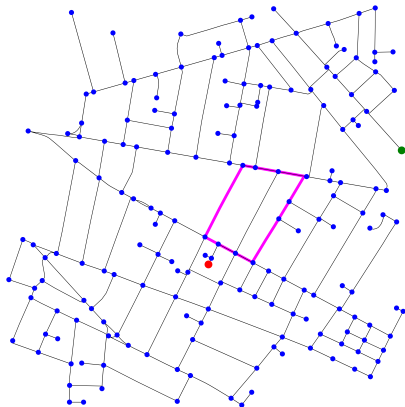


Figure: Edges to measure (in magenta) for $\varepsilon = 7\%$.

Data Needed as a Function of Uncertainty

- $c \in \mathcal{C}_\varepsilon := \{c \in \mathbb{R}^d, (1 - \varepsilon)c_0 \leq c \leq (1 + \varepsilon)c_0\}$.
- More uncertainty ($\varepsilon \uparrow$) $\implies$ more data needed.

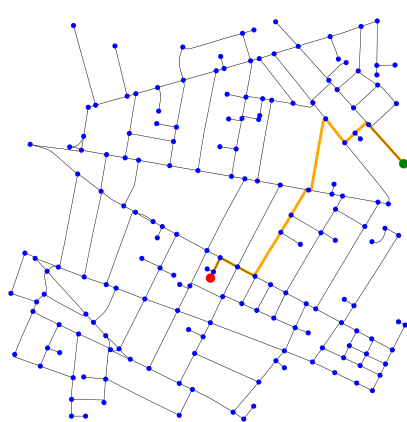


Figure: Shortest path for $c = c_0$.

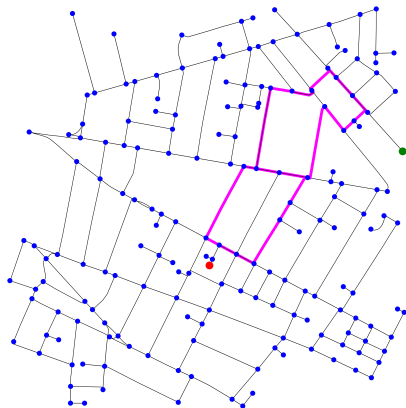


Figure: Edges to measure (in magenta) for $\varepsilon = 7.7\%$.

Data Needed as a Function of Uncertainty

- $c \in \mathcal{C}_\varepsilon := \{c \in \mathbb{R}^d, (1 - \varepsilon)c_0 \leq c \leq (1 + \varepsilon)c_0\}$.
- More uncertainty ($\varepsilon \uparrow$) $\implies$ more data needed.

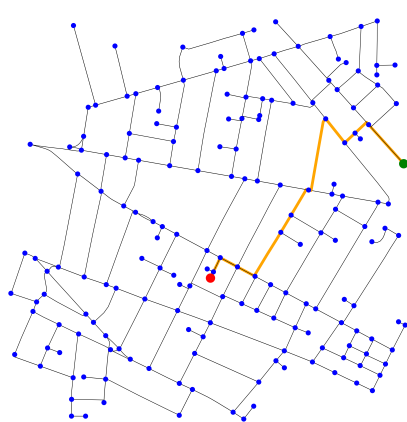


Figure: Shortest path for $c = c_0$.

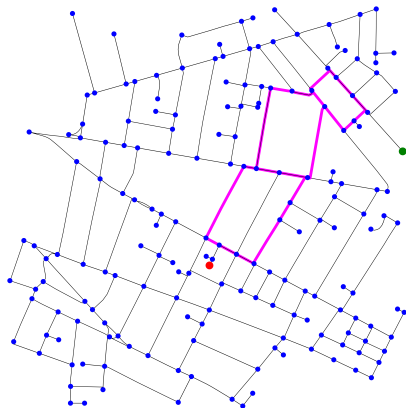


Figure: Edges to measure (in magenta) for $\varepsilon = 10\%$.

Data Needed as a Function of Uncertainty

- More uncertainty ($\varepsilon \uparrow$) $\implies$ more data needed.

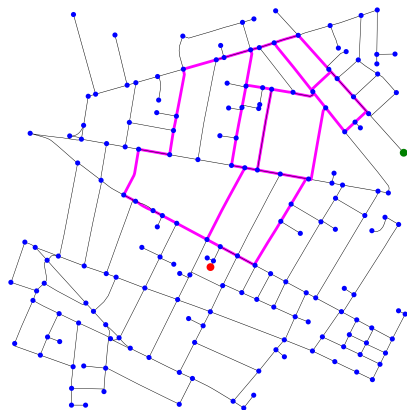


Figure: Edges to measure (in magenta) for $\varepsilon = 30\%$.

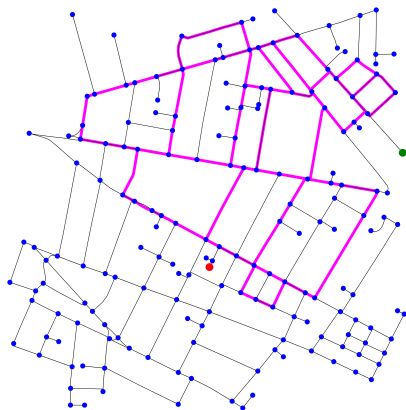


Figure: Edges to measure (in magenta) for $\varepsilon = 60\%$.

Data Needed as a Function of Uncertainty

- When $\varepsilon = 99\%$ ($\sim \mathcal{C} = \mathbb{R}_{\geq 0}^d$), every edge that can induce some optimal solution needs to be measured.

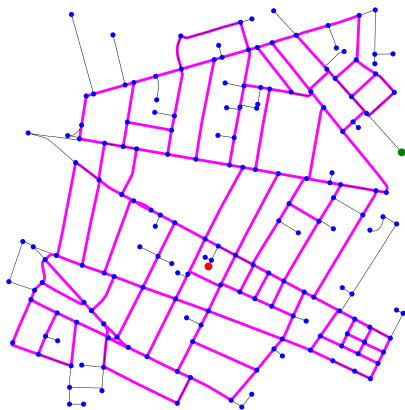


Figure: Edges to measure (in magenta) for $\varepsilon = 99\%$.

Hardness of Constructing Minimum-Size SDDs

Finding Minimum-Size SDDs Is Hard

- Take the clean polyhedral setting: $\mathcal{X}$ and $\mathcal{C}$ are H -polytopes.[†]
- Recall the exact characterization for open convex $\mathcal{C}$:

$$\mathcal{D} \text{ is sufficient} \iff \Delta(\mathcal{X}, \mathcal{C}) \subseteq \text{span } \mathcal{D}, \quad d^* = \dim(\text{dir}(\mathcal{X}^*(\mathcal{C}))).$$

- Worst-case barrier [Ye 26, our work]:
 - computing the intrinsic dimension d^* is NP-hard;
 - deciding global sufficiency of a dataset is coNP-hard.
- Thus a minimum-size *global* SDD is not a tractable target in general: the difficulty is intrinsic, not only an artifact of the MIP formulation above.

[†]An H -polytope is a bounded intersection of finitely many halfspaces, e.g., $P = \{z : Hz \leq h\}$.

NP-Hardness Proof Idea I: Source Problem

Source problem: $3\text{-SAT} \leq_p \text{PISPP-W+}$

PISPP-W+ = *partial inverse shortest path with only weight increases*: given a DAG $G = (V, A)$, initial lengths d , modification costs κ , budget B , and one required arc r , decide whether

$$\exists w \geq 0, \quad \kappa^\top w \leq B, \quad \text{s.t. some shortest } s\text{-}t \text{ path under } d + w \text{ uses } r.$$

We use hard PISPP-W+ instances obtained by the reduction $3\text{-SAT} \leq_p \text{PISPP-W+}$ on DAGs. [Ley-Merkert 25, Thm. 3.1]

- **Represent shortest paths as an LP.** Let $\mathcal{X}$ be the s - t unit-flow polytope of the DAG. Its extreme points are exactly s - t path-incidence vectors. Define the face forbidding r :

$$\mathcal{X}_r := \{x \in \mathcal{X} : x_r = 0\}, \quad \mathcal{X}_r^*(\mathcal{C}) := \text{reachable optima in } \mathcal{X}_r.$$

The reduction asks whether allowing paths that use r adds a new decision-relevant direction.

NP-Hardness Proof Idea II: Reduction

- **Use the path-length structure of the source problem.** All s - t paths avoiding r have baseline length L , while paths using r have baseline length $L + 1$. Thus, at $d \in \mathcal{C}$, all avoiding- r paths are shortest:

$$\mathcal{X}_r^*(\mathcal{C}) = \mathcal{X}_r^< \subseteq \mathcal{X}^*(\mathcal{C}).$$

- **The PISPP-W+ answer becomes a dimension comparison.**

YES instance. There exist $c^+ \in \mathcal{C}$ and $x^+ \in \mathcal{X}^*(c^+)$ with $x_r^+ = 1$, adding one new r -direction:

$$\dim \text{dir}(\mathcal{X}^*(\mathcal{C})) = \dim \text{dir}(\mathcal{X}_r^*(\mathcal{C})) + 1.$$

NO instance. No reachable shortest path uses r , so every reachable optimum avoids r :

$$\begin{aligned} \mathcal{X}^*(\mathcal{C}) &= \mathcal{X}_r^*(\mathcal{C}), \\ \dim \text{dir}(\mathcal{X}^*(\mathcal{C})) &= \dim \text{dir}(\mathcal{X}_r^*(\mathcal{C})). \end{aligned}$$

- **Reduction.** The instances $(\mathcal{X}, \mathcal{C})$ and $(\mathcal{X}_r, \mathcal{C})$ are built in polynomial time. An oracle DStar, where $\text{DStar}(\mathcal{Y}, \mathcal{C}) := \dim \text{dir}(\mathcal{Y}^*(\mathcal{C}))$, decides PISPP-W+ with two queries by checking whether $\dim \text{dir}(\mathcal{X}^*(\mathcal{C}))$ is larger than $\dim \text{dir}(\mathcal{X}_r^*(\mathcal{C}))$. Hence $\text{PISPP-W+} \leq_T^p \text{DStar}$, so computing d^* is NP-hard under polynomial-time Turing reductions.

Beyond Worst-Case Analysis: Learning from Distributional Instances

Beyond Worst-Case: Learning from Distributional Data

- The same LP structure is solved repeatedly, while costs vary across instances:

$$c \sim P_c, \quad c_1, \dots, c_n \stackrel{\text{i.i.d.}}{\sim} P_c, \quad \text{supp}(P_c) \subseteq \mathcal{C}.$$

- Goal: learn a small dataset $\hat{\mathcal{D}}$ that is sufficient for a fresh draw with high probability, rather than for every worst-case $c \in \mathcal{C}$.
- **Traffic example.** The network is fixed and each road-segment travel time lies in a known range. Realized travel times change with time, weather, incidents, and congestion. Historical samples may reveal a few critical road-segment observations that determine the optimal decision for most future instances.
- This is a beyond-worst-case, distributional viewpoint: hard regions of $\mathcal{C}$ may be negligible under P_c .
[Roughgarden 19/21; Balcan 21]

Pointwise SDD vs. Global SDD

For a query dataset $\mathcal{D}$, define

$$\underbrace{s_{\mathcal{D}}(c) := (q^{\top} c)_{q \in \mathcal{D}}}_{\text{measurement observation}}, \quad \underbrace{F_{\mathcal{D}}(c) := \{c' \in \mathcal{C} : s_{\mathcal{D}}(c') = s_{\mathcal{D}}(c)\}}_{\text{data-consistent fiber}}.$$

Pointwise SDD at c . $\mathcal{D}$ is pointwise sufficient at c if

$$\exists x^* \in \mathcal{X}^*(c) \quad \text{s.t.} \quad F_{\mathcal{D}}(c) \subseteq \Lambda(x^*).$$

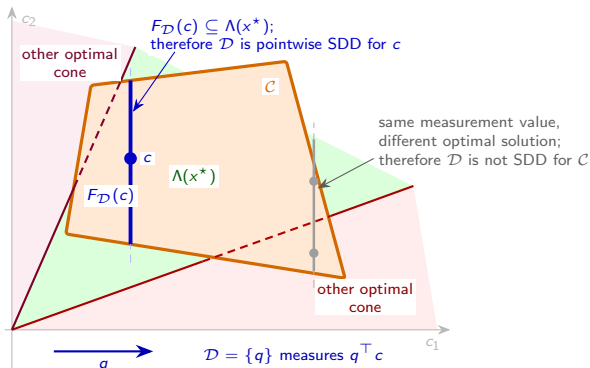
Equivalently, $x^* \in \mathcal{X}^*(c')$ for every cost $c' \in F_{\mathcal{D}}(c)$.

- A **global SDD** must certify every measurement fiber inside $\mathcal{C}$: for each $c' \in \mathcal{C}$, there exists $x^*(c') \in \mathcal{X}^*(c')$ with $F_{\mathcal{D}}(c') \subseteq \Lambda(x^*(c'))$.
- A **pointwise SDD at c** only needs to certify the realized fiber $F_{\mathcal{D}}(c)$. For this realized c , the measurements $s_{\mathcal{D}}(c)$ are enough to find and recover one optimal solution $x^* \in \mathcal{X}^*(c)$; the recovery rule appears later.

Pointwise SDD vs. Global SDD

$$\underbrace{s_{\mathcal{D}}(c) := (q^{\top} c)_{q \in \mathcal{D}}}_{\text{measurement observation}}$$

$$\underbrace{F_{\mathcal{D}}(c) := \{c' \in \mathcal{C} : s_{\mathcal{D}}(c') = s_{\mathcal{D}}(c)\}}_{\text{data-consistent fiber}}$$



In the figure, $\mathcal{D} = \{q\}$ is **not** a global SDD for $\mathcal{C}$.

But $\mathcal{D} = \{q\}$ is **pointwise sufficient at c** , because

$$F_{\mathcal{D}}(c) \subseteq \Lambda(x^*).$$

Pointwise Verification: Hardness and a Tractable Regime

- Verifying pointwise sufficiency at c is equivalent to the cone-containment test

$$\exists x^* \in \mathcal{X}^*(c) \quad \text{s.t.} \quad F_{\mathcal{D}}(c) \subseteq \Lambda(x^*). \quad (\text{CC})$$

- In general, this verification problem is coNP-hard. [Ye 26, our work]
- Reason: H -in- V polytope containment reduces to the test (CC); this containment problem is coNP-hard by Freund–Orlin. [Freund–Orlin 85]
- The same H -in- V construction gives: deciding whether $\mathcal{D} = \emptyset$ is a global SDD is coNP-hard. Since this is a special case, testing whether a general $\mathcal{D}$ is a global SDD is coNP-hard too.
- Under LP nondegeneracy, an optimal basis B gives the simple H -description

$$\Lambda(B) = \{c' : (c')^\top \delta(B, j) \geq 0 \quad \forall j \in N\},$$

where $\delta(B, j)$ is the edge direction obtained by increasing the nonbasic variable j from basis B . Thus (CC) can be checked by polynomially many convex programs.

Constructing Pointwise SDDs

Algorithm 1: Cutting planes for pointwise SDDs. **Input:** LP data, prior $\mathcal{C}$, realized cost c , initial dataset $\mathcal{D}_{\text{init}}$. **Initialize:** $k = 0$, $\mathcal{D}_0 \leftarrow \mathcal{D}_{\text{init}}$, observe $s_{\mathcal{D}_0}(c)$, and solve once at c to obtain an optimal basis B and decision $x(B)$.

- 1 **Form the current fiber.** $F_k := \{c' \in \mathcal{C} : q^\top c' = q^\top c \ \forall q \in \mathcal{D}_k\}$.
- 2 **Test containment in the fixed optimality cone.** Under nondegeneracy, the cone of the initialized basis is

$$\Lambda(B) = \{c' : (c')^\top \delta(B, j) \geq 0 \ \forall j \in N\}.$$

For each $j \in N$, solve the face-intersection subproblem

$$m_j := \min_{c' \in F_k} (c')^\top \delta(B, j). \quad (\text{FI}_j)$$

- 3 **Stop if certified.** If $m_j \geq 0$ for all $j \in N$, then $F_k \subseteq \Lambda(B)$; return $\mathcal{D}_k$ and $x(B)$.
- 4 **Otherwise cut and repeat.** Choose a minimizer $c^{\text{out}} \in F_k$ from a violated test. Let j^* be the first facet of $\Lambda(B)$ crossed by $[c, c^{\text{out}}]$. Query $q_{k+1} := \delta(B, j^*)$, observe $q_{k+1}^\top c$, update $\mathcal{D}_{k+1} := \mathcal{D}_k \cup \{q_{k+1}\}$, set $k \leftarrow k + 1$, and return to Step 1.

Polynomial Implementation and Solution Recovery

- **Polynomial implementation.** Under LP nondegeneracy, solve one initial LP over $\mathcal{X}$ at the realized cost c . If $\mathcal{C}$ is an H -polytope or ellipsoid, each subsequent iteration only performs $d - m$ face-intersection checks: these checks are LPs for H -polytopes and closed form for ellipsoids.
- **Termination and complexity.** Each cut adds a linearly independent direction in $W_\star := \text{dir}(\mathcal{X}^\star(\mathcal{C}))$. Since $\dim(W_\star) = d^\star$, Algorithm 1 adds at most $d^\star$ queries and uses at most $(d^\star + 1)(d - m)$ face-intersection checks.

Decode from a pointwise SDD. Suppose we observe $s_{\mathcal{D}}(c)$ and $\mathcal{D}$ is pointwise sufficient at c . For any $\tilde{c} \in \text{ri } F_{\mathcal{D}}(c)$,

$$\mathcal{X}^\star(\tilde{c}) = \bigcap_{c' \in F_{\mathcal{D}}(c)} \mathcal{X}^\star(c').$$

Thus solving $\min_{x \in \mathcal{X}} \tilde{c}^\top x$ returns a decision optimal for c .

Learning SDDs over Samples

Define the 0–1 sufficiency loss and risk:

$$\ell(\mathcal{D}, c) := \mathbf{1}\{\mathcal{D} \text{ fails at } c\}, \quad R(\mathcal{D}) := \Pr_{c \sim P_c} [\ell(\mathcal{D}, c) = 1].$$

Algorithm 2: Learning sufficient decision datasets over samples.

Input: prior $\mathcal{C}$, LP data, samples $c_1, \dots, c_n$.

Initialize: $\mathcal{D}_0 = \emptyset$, hard set $T = \emptyset$.

For $i = 1, \dots, n$:

- ① Run Algorithm 1 on c_i , warm-started with $\mathcal{D}_{i-1}$.
- ② Let the returned dataset be $\mathcal{D}_i$.
- ③ If $\mathcal{D}_i \neq \mathcal{D}_{i-1}$, mark i hard: $T \leftarrow T \cup \{i\}$.

Return: $\mathcal{D}_n$ and the hard subsequence indexed by T .

- Samples with $\mathcal{D}_i = \mathcal{D}_{i-1}$ are already certified by the current dataset.
- **Hard samples are exactly the elements of T :** only they add new decision-relevant directions and induce the stable compression scheme next.

Stable Sample-Compression Scheme

Algorithm 2 induces a realizable stable sample-compression scheme.

[Hanneke–Kontorovich 21]

Three key properties

- **Realizability:** $\ell(\mathcal{D}_n, c_i) = 0$ for every training sample $i = 1, \dots, n$.
 - **Stability:** $\mathcal{D}_n$ is fully determined by the hard subsequence $(c_i)_{i \in T}$; deleting non-hard samples changes nothing.
 - **Small compression size:** $|T| \leq |\mathcal{D}_n| \leq d^*$.
-
- $|T| \leq |\mathcal{D}_n|$ holds because each hard sample adds at least one query direction.
 - $|\mathcal{D}_n| \leq d^*$ holds because Algorithm 1 only adds independent directions in W_* .
 - This stable compression structure leads directly to the distribution-free PAC certificate next.

Distribution-Free PAC Certificate

Theorem (Certificate via stable compression)

For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $c_1, \dots, c_n$, Algorithm 2 returns $(\mathcal{D}_n, T)$ satisfying

$$R(\mathcal{D}_n) \leq \frac{4}{n} \left(6|T| + \ln \frac{e}{\delta} \right) \leq \frac{4}{n} \left(6d^* + \ln \frac{e}{\delta} \right).$$

- Fast rate:

$$R(\mathcal{D}_n) = O\left(\frac{d^* + \ln(1/\delta)}{n}\right).$$

- For Algorithm 2, the d^*/n dependence is tight up to logarithmic factors.

[Ye 26, our work]

Comparison to Data-Driven Algorithm Design

- **Data-driven algorithm design** is a line of work led by Nina Balcan and her group. It starts from a parameterized algorithm family $\{A_\theta\}$ and training instances drawn from an unknown application-specific distribution. [COLT 17; ICML 18; FOCS 18; STOC 21/JACM 24; NeurIPS 24]
- It chooses a parameter $\hat{\theta}$ with strong average training performance, and proves generalization guarantees for the future performance of the learned algorithm $A_{\hat{\theta}}$.
- **Data-driven projection for LPs** learns a low-dimensional projection and proves generalization bounds for the objective-value gap of the learned projection; the empirical projection-learning problem is generally nonconvex. [Sakaue–Oki 24]
- These works control *performance gaps* for the learned object ($A_{\hat{\theta}}$ or a projection), typically at the uniform-convergence scale $\tilde{O}(1/\sqrt{n})$.

Our contrast. We learn a measurement set $\mathcal{D}$, not a parameter setting or a projection matrix. The loss is exact decision-recovery failure: $\ell(\mathcal{D}, c) = \mathbf{1}\{\mathcal{D} \text{ fails at } c\}$.

Algorithm 2 constructs $\mathcal{D}$ in polynomial time under our tractability conditions, and realizable stable compression gives $R(\mathcal{D}_n) = \tilde{O}(d^*/n)$.

Intuition for the Fast Rate

- A standard $\tilde{O}(1/\sqrt{n})$ bound is the standard deviation scale for estimating an unknown Bernoulli error rate:

$$\text{sd}(\hat{R}_n) \approx \sqrt{R(1-R)/n}.$$

Without realizability, this is the natural scale.

- In a realizable stable compression scheme, the learner returns h_S with zero training loss, and h_S is determined by a small compression set $T(S) \subseteq S$.

$$|T(S)| \leq k, \quad \ell(h_S, z_i) = 0 \quad (i \leq n).$$

- The risk is *missing mass*, not estimation noise: a fresh test example fails only if it would become one of the hard examples selected by the compression rule.
- Compression intuition.** Draw $S^+ = (z_1, \dots, z_{n+1})$ and hold out a uniformly random index J . Run the compression rule on S^+ . If $J \notin T(S^+)$, stability says removing z_J does not change the learned hypothesis, and the full run has zero loss on z_J . Hence

$$\Pr\{\text{test fails}\} \lesssim \frac{|T(S^+)|}{n+1} \leq \frac{k}{n+1}.$$

For our SDD learner, $k = d^*$, so $R(\mathcal{D}_n) = \tilde{O}(d^*/n)$.

[Hanneke–Kontorovich 21]

Application: Contextual Linear Optimization

- Data are drawn as $(\xi, c) \sim P$, where $\xi \in \mathbb{R}^p$ is context and $c \in \mathcal{C} \subseteq \mathbb{R}^d$ is the downstream LP cost.
- A predictor $f(\xi)$ induces the plug-in decision

$$x^*(f(\xi)) \in \arg \min_{x \in \mathcal{X}} f(\xi)^\top x.$$

- The **Smart Predict-then-Optimize (SPO)** loss measures the extra cost of using the predicted cost $\hat{c}$:

$$\ell_{\text{SPO}}(\hat{c}, c) := c^\top x^*(\hat{c}) - c^\top x^*(c) \geq 0.$$

- SPO is decision-aware: prediction errors matter only through the downstream decision.
- If decisions only depend on $W_\star$, train a coordinate predictor in $\mathbb{R}^{d^*}$ and lift it back to the original cost space. [Elmachtoub–Grigas 22; El Balghiti et al. 23]

Improved SPO Generalization Bound

Stage I: representation

Run Algorithm 2 over cost samples to learn a decision-relevant subspace

$$\widehat{W} = \text{span } \widehat{\mathcal{D}} \approx W_*$$

Stable compression gives a pointwise-sufficiency failure certificate of order $O(d^*/n)$.

Stage II: prediction

Train an SPO predictor in the learned coordinates, then lift the prediction to $\mathbb{R}^d$ before solving the LP.

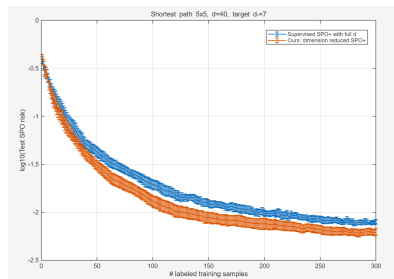
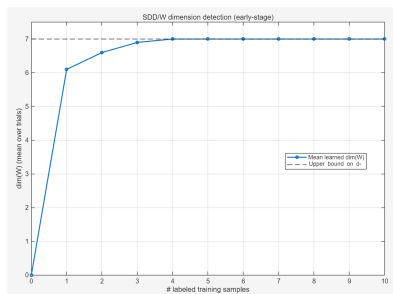
$$g(\xi) \in \mathbb{R}^{d^*} \mapsto \hat{c}(\xi) \in \mathbb{R}^d.$$

Theorem (Compressed SPO generalization [El Balghiti et al. 23; Ye 26])

For linear coordinate predictors, the SPO generalization term scales as

$$\tilde{O}\left(\sqrt{\frac{d^* p}{n}}\right) \text{ instead of } \tilde{O}\left(\sqrt{\frac{dp}{n}}\right).$$

Numerical Illustration: Shortest Path



- 5-by-5 shortest-path grid: ambient cost dimension $d = 40$, context dimension $p = 5$, intrinsic dimension $d^* = 7$.
- Stage I learns a low-dimensional decision-relevant subspace.
- Stage II trains SPO predictors in that subspace, improving out-of-sample SPO risk relative to full-dimensional training.

Open Directions

- **Hardness beyond degeneracy and approximability.** Do the hardness results persist under nondegeneracy assumptions on $\mathcal{X}$? Can we compute a global SDD whose cardinality is within a constant factor of the minimum?
- **Adaptive decision information.** What is the worst-case adaptive query complexity for pointwise sufficiency? Can test-time branching use fewer than d^* queries per instance?
- **Extension to noisy and robust settings.** Our framework assumes exact measurements $q_i^\top c$. With noisy observations, one may need margin conditions controlling how often c lies near cone boundaries. Can robust sample-compression schemes handle corrupted samples or adversarial outliers (e.g., an ε -fraction of arbitrary corruptions)?
- **SDD geometry beyond LPs.** Extend decision-sufficient representations to mixed-integer programs, quadratic programs, or LPs with varying constraints.